Secure Corporate Network Project

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Lab Environment: Cisco Packet Tracer
Project Type: Defensive Network Security Architecture

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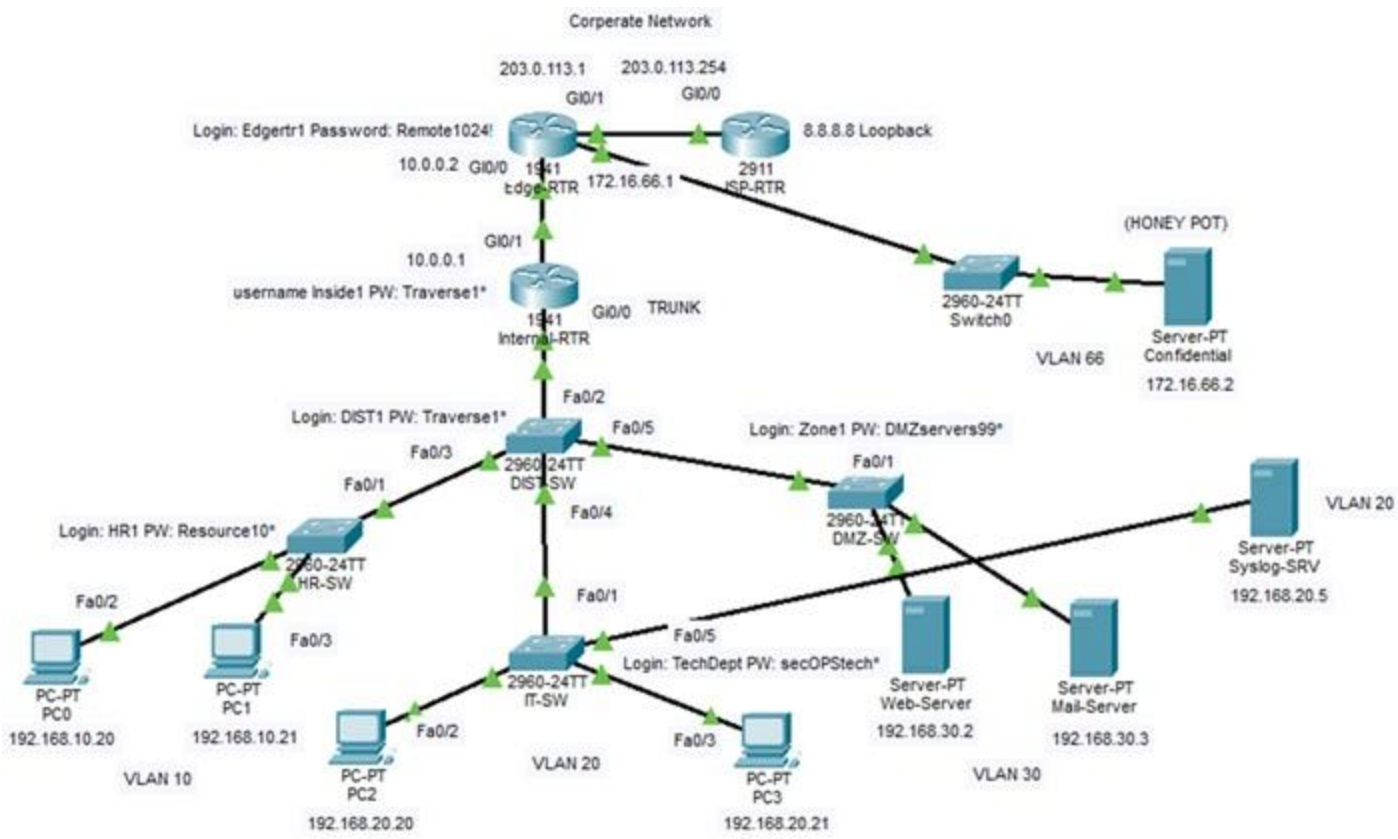
1. Executive Summary

Purpose

This project demonstrates the design, implementation, and validation of a secure enterprise network while explicitly showcasing a practical understanding of the network infrastructure, traffic flows, and security controls.

Key Objectives

- Demonstrate in-depth knowledge of enterprise network architecture and data flow
- Implement logical network segmentation using VLANs to reduce attack surface
- Enforce controlled inter-VLAN communication through access control lists (ACLs)
- Provide secure Internet access using NAT and PAT while maintaining traffic visibility
- Centralize security and operational logging using Syslog functioning as a SIEM platform
- Secure network device management through SSH and AAA authentication
- Isolate public-facing resources within a hardened DMZ and deploy a honeypot to observe and analyze malicious activity



2. IP Addressing & VLAN Design

VLAN Structure

VLAN	Purpose	Subnet
10	HR Users	192.168.10.0/24
20	IT / Management	192.168.20.0/24
30	DMZ Servers	192.168.30.0/24

Key Hosts

- Syslog Server: 192.168.20.5
- Web Server (DMZ): 192.168.30.2
- Mail Server (DMZ): 192.168.30.3

4. Security Controls Implemented

4.1 Access Control Lists (ACLs)

Objective: Restrict unauthorized lateral movement.

Why This Control Matters

The purpose of restricting unauthorized lateral movement is to prevent a compromise in one VLAN from spreading into more sensitive network zones.

In enterprise environments, attackers rarely stop at the first compromised endpoint. After gaining access to a user workstation, they often attempt to move laterally across the network to reach:

- administrative systems
- internal file shares
- authentication services
- public-facing servers in the DMZ
- logging or monitoring infrastructure

By applying ACLs between VLANs, the network enforces **least-privilege communication paths**, ensuring each segment can only reach the resources required for its business function.

For example, **HR workstations in VLAN 10 have no legitimate reason to directly access DMZ servers in VLAN 30**, so blocking that path reduces attack surface and helps contain breaches.

This control limits:

- ransomware spread
- credential theft pivoting
- unauthorized reconnaissance
- privilege escalation attempts
- malware propagation between departments

In this project, ACL enforcement ensures that even if an HR endpoint is compromised, the attacker cannot directly pivot into public-facing servers or management infrastructure.

Examples:

- HR VLAN blocked from DMZ
- DMZ blocked from Internet egress

- Explicit permit rules for required traffic only

Verification:

- ICMP tests

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::240:BFF:FEE5:6096
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.10.20
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.10.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ACL configured to deny VLAN 10 access to VLAN 30
```

- ACL hit counters

```
INTERNAL-RTR#show ip access-lists HR-BLOCK-DMZ
Extended IP access list HR-BLOCK-DMZ
    deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255 (4 match(es))
    permit ip any any
```

4.2 NAT / PAT Configuration

Objective: Enable secure outbound Internet access while hiding internal IPs.

- PAT using public IP 203.0.113.1
- Internal subnets translated dynamically

Why This Matters

This control is important because enterprise endpoints require Internet connectivity for legitimate business operations such as software updates, cloud access, web browsing, and security telemetry, but exposing internal private addresses directly to the Internet would significantly increase risk.

NAT/PAT provides secure outbound connectivity by translating many internal systems into a single public IP address, which preserves routability while concealing the organization's internal addressing scheme from external networks.

This matters because it:

- reduces external visibility of internal hosts
- makes internal reconnaissance more difficult for attackers
- limits direct inbound targeting of workstations
- conserves public IPv4 addresses
- centralizes outbound traffic at a controlled egress point
- improves logging and traffic inspection opportunities

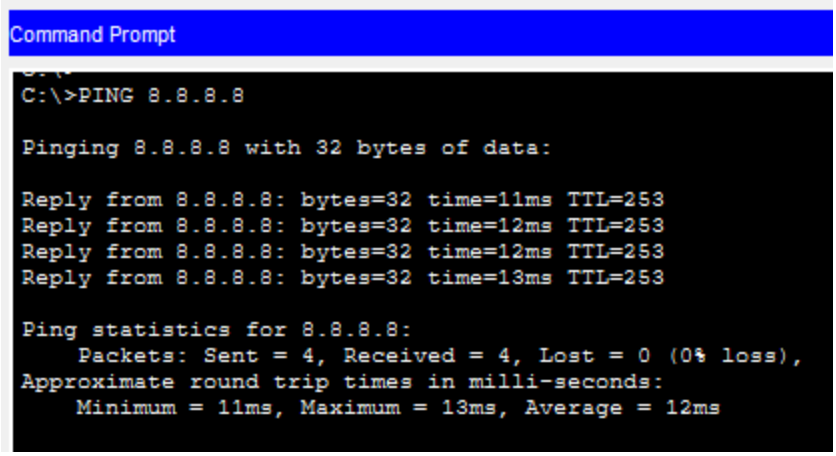
From a defensive perspective, PAT also helps security teams quickly trace outbound sessions back to internal systems through translation tables and firewall logs, which is critical during incident response.

In this project, successful translation of HR VLAN traffic to 203.0.113.1 demonstrates that internal users can securely access external resources while keeping private addressing hidden from public networks.

Verification:

- Show IP NAT translations
- Successful ping to 8.8.8.8

```
EDGE-RTR#show ip nat translations
show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 203.0.113.1:5     192.168.10.20:5    8.8.8.8:5          8.8.8.8:5
icmp 203.0.113.1:6     192.168.10.20:6    8.8.8.8:6          8.8.8.8:6
icmp 203.0.113.1:7     192.168.10.20:7    8.8.8.8:7          8.8.8.8:7
icmp 203.0.113.1:8     192.168.10.20:8    8.8.8.8:8          8.8.8.8:8
```



4.3 Centralized Logging (Syslog / SIEM)

Objective: Monitor security events across network devices.

Why This Matters
Centralized logging is critical because preventive controls such as ACLs, NAT, and AAA reduce risk but do not provide full visibility into attacker behavior or system misuse. Security teams need a reliable telemetry source to detect suspicious activity, validate control effectiveness, and reconstruct events during investigations.

By forwarding logs from routers and switches into a centralized Syslog server, the network creates a SIEM-like visibility layer that captures authentication events, ACL denies, interface state changes, and configuration modifications. This allows defenders to quickly identify brute-force attempts, unauthorized configuration changes, lateral movement attempts, or service disruptions.

In enterprise environments, centralized logging significantly improves incident response by enabling analysts to correlate events across multiple devices, establish attack timelines, and verify whether security controls performed as expected.

In this project, Syslog demonstrates how blue teams gain actionable visibility into network activity, transforming raw device events into usable security telemetry for threat detection and response.

Logged Events:

- Login success/failure
- ACL denies
- Configuration changes
- Interface state changes

SIEM Role:

- Syslog server simulates a centralized SIEM solution

Syslog

Syslog

Service On Off

	Time	HostName	Message
1	03.02.1993 11:03:18.485 PM	10.0.0.2	%SEC_LOGIN-5-LOGIN_FAILED: Login failed [user: Edgertr1] [Source: 0.0.0.0] [localport: 0] [Reason: Login Authentication Failed] at 23:03:18 UT...
2	03.02.1993 11:03:48.229 PM	10.0.0.2	%SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: Edgertr1] [Source: 0.0.0.0] [localport: 0] at 23:03:48 UTC Tue Mar 2 1993...
3	03.02.1993 11:08:09.416 PM	10.0.0.2	%SYS-5-CONFIG_I: Configured from console by console
4	03.02.1993 11:17:35.427 PM	10.0.0.2	%SYS-5-CONFIG_I: Configured from console by console
5	03.04.1993 01:38:05.239 AM	192.168.20.1	%SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: Inside1] [Source: 0.0.0.0] [localport: 0] at 01:38:05 UTC Thu Mar 4 1993...
6	03.02.1993 11:22:40.602 PM	10.0.0.2	%SYS-5-CONFIG_I: Configured from console by console
7	03.04.1993 12:01:20.961 AM	192.168.20.10	%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
8	03.04.1993 12:01:20.961 AM	192.168.20.10	%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, ...
9	03.04.1993 12:03:40.065 AM	192.168.20.10	%LINK-3-UPDOWN: Interface FastEthernet0/5, changed state to down
10	03.04.1993 12:03:40.065 AM	192.168.20.10	%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, ...
11	03.04.1993 12:03:49.816 AM	192.168.20.10	%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
12	03.04.1993 12:03:49.816 AM	192.168.20.10	%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, ...

4.4 Secure Management (AAA & SSH)

Objective: Prevent unauthorized device access.

Why This Matters
Securing administrative access to network infrastructure is critical because routers and switches represent high-value targets in enterprise environments. If an attacker gains unauthorized management access, they can modify routing paths, disable security controls, intercept traffic, or create persistence mechanisms that impact the entire network.

SSH protects management sessions by encrypting credentials and command traffic in transit, preventing interception through packet sniffing or man-in-the-middle attacks. Disabling Telnet removes the risk of plaintext credential exposure, which is a common weakness in poorly secured legacy environments.

AAA strengthens this control by enforcing authenticated administrative access, ensuring only approved users can manage infrastructure devices. It also creates accountability by tying access attempts to named user credentials, which improves auditing and incident investigations.

In enterprise security operations, these controls reduce the risk of credential theft, unauthorized configuration changes, and insider misuse while improving visibility into privileged access activity.

In this project, AAA and SSH demonstrate a layered management-plane defense strategy where confidentiality, access control, and administrative accountability are enforced across the network.

- SSH enabled on routers and switches
- Telnet disabled

```
HR-SW#telnet 192.168.10.10
Trying 192.168.10.10 ...Open

[Connection to 192.168.10.10 closed by foreign host]
HR-SW#
```

- Local AAA authentication

```
*********************************************************
* WARNING: This is a secured corporate network system.   *
* Unauthorized access is strictly prohibited.             *
* All activities are monitored and logged.                *
*********************************************************


User Access Verification

Username: Techdept
Password:
IT-SW>en
Password:
IT-SW#
```

Login to IT-SW is configured as Username: Techdept with Password:secOPStech*

(The Credentials detailed would be redacted in a real work environment)

4.5 DMZ Isolation

Objective: Protect internal network from public services.

Why This Matters
Public-facing services such as web and mail servers are among the most frequently targeted assets in enterprise environments because they must remain reachable from less trusted networks. If these systems are compromised, attackers often attempt to pivot deeper into internal infrastructure to access user systems, management devices, or sensitive business data.

A DMZ reduces this risk by placing exposed services into an isolated security zone separated from internal user and management VLANs. This segmentation limits the blast radius of a compromise by ensuring public services cannot directly communicate with sensitive internal resources unless explicitly required by policy.

In this project, one-way communication controls and explicit Internet egress restrictions demonstrate a containment-first security strategy. Even if a DMZ server is breached, the attacker’s ability to move laterally, establish outbound command-and-control channels, or pivot into internal systems is significantly reduced.

From a blue-team perspective, DMZ isolation improves incident response by clearly defining trust boundaries, simplifying forensic scoping, and reducing the number of systems potentially affected during a public-service compromise.

This control demonstrates enterprise-grade segmentation principles focused on attack surface reduction, breach containment, and preservation of internal network integrity.

- Public servers placed in VLAN 30
- One-way communication controls
- Internet access explicitly blocked

Verification:

- DMZ → Internet ping fails

```
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

- DMZ → Internal gateway succeeds

```
DMZ-SW>ping 10.0.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/1/3 ms
```

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4.6 Honeypot Deployment

Objective: Detect reconnaissance or malicious activity.

Why This Matters
Traditional preventive controls focus on blocking malicious activity, but defenders also benefit

from intentionally exposing controlled deception assets that attract unauthorized interaction. A honeypot serves this purpose by acting as a monitored decoy system designed to detect reconnaissance, scanning, exploitation attempts, or lateral movement activity that should never occur during normal business operations.

Because legitimate users and production systems have no operational reason to interact with the honeypot, any connection attempt is treated as a high-confidence security signal. This dramatically reduces false positives and allows defenders to quickly identify suspicious behavior such as port scanning, service enumeration, brute-force attempts, or unauthorized probing of exposed services.

In this project, the honeypot is deployed on a dedicated subnet off the edge router with strict ACL isolation to ensure that malicious interaction can be safely observed without creating risk to the internal enterprise environment. All connection attempts are logged, providing defenders with actionable intelligence on attacker tactics, techniques, and procedures (TTPs).

From a blue-team perspective, this control improves early threat detection, supports threat hunting, and provides valuable forensic insight into how attackers enumerate and target exposed systems.

This demonstrates a proactive security strategy centered on deception, telemetry generation, and adversary behavior analysis.

- Dedicated subnet off Edge Router
- Strict ACL isolation
- Logging of all interaction attempts

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5. Testing & Validation

Test	Expected Result	Outcome
HR → DMZ Ping	Blocked	Pass
HR → Internet	Allowed	Pass
DMZ → Internet	Blocked	Pass
SSH Login Failure	Logged	Pass
NAT Translation	Visible	Pass

Why This Matters

Security controls only provide value when their effectiveness is verified through repeatable testing and observable outcomes. Validation ensures that implemented protections behave as intended under realistic traffic and access scenarios rather than relying solely on configuration assumptions.

In this project, each control was tested against an expected defensive outcome, including segmentation enforcement, Internet access policies, logging visibility, and NAT translation behavior. Successful test results confirm that preventive, detective, and containment controls are functioning together as an integrated security architecture.

From a blue-team perspective, validation mirrors real-world control assurance processes such as segmentation audits, firewall rule verification, privileged access monitoring, and incident response readiness checks. These tests demonstrate that security controls are not only configured, but actively enforced and observable.

This testing methodology highlights a security engineering approach focused on **measurable outcomes, control assurance, and evidence-based verification**, which are critical in enterprise environments and security operations roles.

7. Tools & Technologies Used

- Cisco Packet Tracer
- Cisco IOS (Routers & Switches)
- ACLs, NAT/PAT, VLANs
- Syslog (SIEM simulation)
- AAA & SSH

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8. Lessons Learned

- Importance of ACL order and placement
- Logging is critical for validation
- Network segmentation drastically reduces attack surface
- Blue-team visibility is as important as prevention

This project reinforced the importance of designing security with both prevention and visibility in mind. Implementing controls such as ACLs, NAT/PAT, centralized logging, and segmentation demonstrated that effective defense is not achieved through a single mechanism, but through layered security strategies working together. Throughout the build and validation process, it became clear that proper control placement, continuous monitoring, and verification are critical to maintaining a secure and resilient network. Beyond configuration, this project emphasized the value of thinking like a defender—anticipating attacker behavior, limiting blast radius, and ensuring that suspicious activity can be detected and investigated. Overall, this project highlights a practical understanding of enterprise network defense, combining secure architecture design with operational visibility and evidence-based validation to support real-world cybersecurity practices.